

Title: EEG Microstate Analysis

Category: Analysis

Libraries MNE, Pycrostates, NumPy, Matplotlib

1. Objective: To segment continuous, chaotic EEG signals into a sequence of distinct, semi-stable spatial patterns called "Microstates." These microstates represent the fundamental building blocks of brain activity and are heavily linked to resting-state functional networks (e.g., Visual, Auditory, Attention). Instead of looking at frequencies (like Spectral Analysis), this method looks at how the brain's electrical landscape changes over time.

2. How it Works (Mathematical Concept & Algorithm): The algorithm uses Modified K-Means (ModKMeans) clustering via the pycrostates library and operates in four main steps:

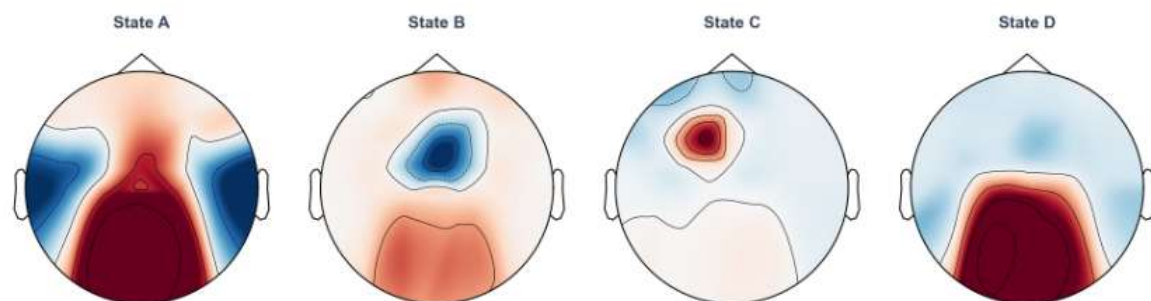
- **Step 1: Signal Preparation & Normalization** The data undergoes Global Z-Score normalization to ensure robust math regardless of the headset's voltage scale. It is then re-referenced to the "Average Reference" so that the sum of all electrical potentials across the head equals zero at any given moment.
- **Step 2: Finding GFP Peaks** The algorithm calculates the Global Field Power (GFP), which is the standard deviation of all electrodes at a given time point. GFP peaks represent moments where the brain's topographic map is strongest and most stable (highest signal-to-noise ratio).
- **Step 3: ModKMeans Clustering** The algorithm extracts the topographic maps only at the GFP peaks and feeds them into a Modified K-Means clustering algorithm. It groups thousands of these peak maps into a small number of representative clusters (default is 4). Polarity is ignored (a positive peak and a negative peak of the same shape are grouped together).
- **Step 4: Backfitting & Smoothing** Once the 4 master maps are identified, the algorithm goes back to the continuous EEG data. It calculates the spatial correlation between every single time point and the 4 master maps, assigning each moment to the "winning" state. A temporal smoothing window (e.g., 5 samples) is applied to remove tiny, biologically impossible flickering between states.

3. The 4 Canonical Microstates (Biological Meaning): The algorithm extracts 4 standard states (A, B, C, D) and calculates their Occurrence (Hz), Mean Duration (ms), and Coverage (%):

- **Microstate A (Auditory Network):** Features a Right-Front to Left-Back diagonal orientation. Linked to phonological processing and the auditory cortex.
- **Microstate B (Visual Network):** Features a Left-Front to Right-Back diagonal orientation. Linked to the visual resting-state network.
- **Microstate C (Salience Network):** Features a strong Anterior-Posterior (Front-to-Back) orientation. Linked to subjective interoception and transitioning between internal/external thoughts.
- **Microstate D (Attention Network):** Features a Frontal/Central focus. Linked to the dorsal attention network and executive function.

4. Visual Example:

Because visualizing spatial data requires knowing *where* the sensors are on the head, this step dynamically changes its output based on your Channel Mapping configuration. Scenario A: Mapped Channels (Topographic Maps) *If you use the Channel Mapping tool to map your raw columns to the standard 10-20 system (e.g., channel_3 mapped to Fz, as seen in your screenshot):* The algorithm successfully plots traditional Topographic Head Maps (Topomaps). Red indicates positive voltage distribution, and blue indicates negative. The spatial distribution clearly shows the A, B, C, and D patterns across the scalp.



Scenario B: Unmapped Channels (Polar Plots) *If you do not map your channels, and the algorithm only receives generic names (e.g., channel_1, channel_2):* The algorithm cannot draw a head because it doesn't know where channel_1 is located. Instead, it triggers a fallback and generates Polar Plots (Circular Bar Charts). Each bar represents a generic channel's contribution to that microstate (Red = Positive contribution, Blue = Negative contribution).

